Selected Literature Pages

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Source	Included pages
Li, "Rescaled-range and power spectrum analyses on well-logging data"	PDF pp. 1–3, 6, 11–12
Schepers et al., "Four methods to estimate the fractal dimension from self-affine signals"	PDF pp. 1–2, 6–8
Hough, "On the use of spectral methods for the determination of fractal dimension"	PDF pp. 1–4
Li and Cheng, "Fractal Correction of Well Logging Curves"	PDF pp. 1–4

Suggested use:

1. Start with Li (2003): it is the most directly relevant well-log paper.
2. Use Schepers for estimator behavior and finite-sample cautions.
3. Use Hough for the warning that a power-law spectrum alone does not prove fractality.
4. Read Li and Cheng only to understand the historical application of local scale-dependent correction; it is not part of the current pilot's executed method.

Boundary: Do not treat any reproduced application result as validation of the current FORCE 2020 pilot. The current MTP result remains the result reported in the first nine pages.

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Rescaled-range and power spectrum analyses on well-logging data

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SUMMARY

Fractal theory and related developments have significantly enlightened our understanding of many natural phenomena and led to an upsurge in research efforts in many different disciplines. This has unfortunately caused some ambiguities, misunderstandings, and even misuses in the published literature concerning some new concepts in fractal theory and their actual implementations. As new applications and ideas arise almost daily, it is important to present the methods in a common framework to minimize further confusion. In this paper, we scrutinize and compare two of the most frequently applied methods for characterizing fractal time-series, namely the rescaled-range (R/S) and the power spectrum method, and describe their proper applications in wire line well-logging data. We outline the problems related to applications of these two methods. Based on numerical analysis on fractional Gaussian noise (fGn) and fractional Brownian motion (fBm), we demonstrate the clear necessity to distinguish between fGn-like and fBm-like data series. For fBm-like data series, the use of their successive increments is proposed against using various other alternatives in R/S analysis in order to make the estimated Hurst exponent (Hu) comparable with the global scaling exponent (H) from power spectrum analysis. We argue that well-logging data are generally analogous to fBm, and thus we use their incremental series rather than raw well data themselves to estimate Hu when applying R/S. We also reveal a connection between the transient zone length in the R/S technique and the shortest-wavelength that can be included for a linear regression in the power spectrum method.

Key words: borehole geophysics, Fourier transforms, fractals, geostatistics, numerical techniques.

1 INTRODUCTION

Careful studies have revealed some problems in the current application of rescaled-range analysis (R/S) and the power spectrum in determining the Hurst exponent (Hu) and the fractal dimension (D). In particular, it appears to be evident that R/S and power spectrum methods rarely give similar estimates of Hu, and in many cases differ significantly if not performed with particular care. Individually, both R/S and the spectrum method have been criticized for their inaccuracies (Fox 1989; Hough 1989; Lo 1991; Bassingthwaighte & Raymond 1994; North & Halliwell 1994). However, discussions on the obvious discrepancies between the two are often safely ignored. It is our view that neglecting this issue could hamper true understanding of Hu and D, and could make less meaningful estimates. This paper, through a detailed mathematical analysis and case studies, is meant to clarify some of the ambiguities involved in their applications and make results from different authors and/or techniques more consistent and comparable.

The concept of the Hurst exponent was first introduced by Hurst (1951) via his well-known empirical relationship, called the rescaled-range technique, in studying long-term discharge variations of the River Nile, and later publicized and solidified by Mandelbrot & Wallis (1969) and Feder (1988). Since then it has gained popularity and been applied in many subjects, including network traffic, bioengineering, physical simulation and of course, geology and geophysics. The key point of this technique is the following empirical relationship:

$$R/S = (a\tau)^{Hu}, \tag{1}$$

where a is a constant, R is the difference between the maximum and minimum accumulated departure from the mean within time span τ, S is the standard deviation serving as a scaling factor and Hu is the Hurst exponent coined by Mandelbrot (1982). The accumulated departure with respect to the mathematical mean $\langle\xi\rangle$ of a time-series ξ_i is

$$X(t) = \sum_{i}^{N}(\xi_i - \langle\xi\rangle), \tag{2}$$

where t is time, i is the time sample index and N is the number of time samples. The range R is defined as

$$R(\tau) = \max_{1\le t\le\tau} X(t) - \min_{1\le t\le\tau} X(t) \tag{3}$$

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and $S(\tau)$ is the standard deviation

$$S(\tau) = \left[\frac{1}{\tau} \sum_{t=1}^{\tau} (\xi_t - \langle \xi \rangle)^2 \right]^{1/2}. \quad (4)$$

Eq. (1) clearly reveals a scaling-invariant property of the data with respect to the period τ. Specifically, the rescaled range R/S is linearly proportional to $a\tau$ on a log–log plot, the slope of which is equal to Hu.

Among various definitions for a fractal time-series, one of the commonly used ones is

$$F(\sigma t) = \sigma^H F(t). \quad (5)$$

It describes a self-affine process, i.e. a subset of a fractal process is a scaled version of the fractal set itself. We call H in eq. (5) the global scaling exponent and σ is a scaling operator. For a self-affine time-series defined by eq. (5), it can be shown that its power spectral density $S(f)$ has a power-law dependence on frequency f:

$$S(f) \propto f^{-\beta}, \quad (6)$$

where we call β the spectrum power. Turcotte (1997) showed that the following relationship between H and β exists:

$$\beta = 2H + 1. \quad (7)$$

We are now confronted with a dilemma as to whether the global scaling exponent (H) derived from eq. (7) is equal to the Hurst exponent (Hu) from the R/S technique. In many studied cases no distinctions are made between H and Hu, and they are often used interchangeably (Hewett 1986; Shiomi *et al.* 1997; Dolan *et al.* 1998; Taqqu & Teverovsky 1998), while in a few situations H is differentiated from Hu (Turcotte 1997). This conflict has posed confusion and misinterpretation in estimating Hu. It has been observed that estimating Hu and H for the same data series, but using two different techniques above, can give significantly different results. This prompts us to consider in what sense H and Hu can be considered equivalent.

2 IS H EQUAL TO Hu?

To answer this question it is helpful to discuss two basic fractal processes that are closely related: the fractional Brownian motion (fBm) and fractional Gaussian noise (fGn). fGn is the successive steps or increments of fBm; in other words, the accumulative summation of fGn gives fBm. While fBm is non-Gaussian and non-stationary, fGn is Gaussian and stationary.

It is straightforward to prove that for fBm with spectrum power $1 \le \beta_{\mathrm{fBm}} \le 3$, the corresponding fGn has $-1 \le \beta_{\mathrm{fGn}} \le 1$. An fBm series $X(t)$ is nowhere differentiable, but in a distribution when the subsequent interval $\Delta X = X(t_i) - X(t_{i-1})$ is sufficiently small compared with the total length of X, from the definition of the derivative we can write

$$G(t) = \Delta X = X(t_i) - X(t_{i-1}) \cong \Delta t X'(t), \quad (8)$$

where Δt is a time sampling interval. Taking the first derivative of $X(t)$ in the time domain is equivalent to multiplying $X(f)$, which is the Fourier transform of $X(t)$, by $-i2\pi\ f$ (Bracewell 1965), $i = (-1)^{1/2}$. By taking the Fourier transform of both sides of (8), we have

$$G(f) \cong \Delta t[-2\pi i f X(f)]. \quad (9)$$

Here we see that the non-differentiability of fBm can actually be relaxed in the frequency domain. The power spectra are given by

$$|G(f)|^2 \cong |\Delta t[-i2\pi f X(f)]|^2 = -4\Delta t^2 \pi^2 f^2 |X(f)|^2. \quad (10)$$

Here $|\ G(f)|^2$ is the power spectrum of the fGn and $|X(f)|^2$ is the power spectrum of the fBm. Because $|X(f)|^2 \propto f^{-\beta}$, we have $|G(f)|^2 \propto f^{-\beta+2} = f^{-(\beta-2)}$. This proves that

$$\beta_{\mathrm{fGn}} \cong \beta_{\mathrm{fBm}} - 2. \quad (11)$$

Or from eq. (7),

$$H_{\mathrm{fGn}} \cong H_{\mathrm{fBm}} - 1. \quad (12)$$

The proof of eqs (11) and (12) reveals that β and H are indicators of the roughness of the data. The larger H is the smoother the data. By taking a first-order derivative we increase the roughness of the data but decrease the value of H by 1. Now eq. (7) can be extended to:

$$\begin{aligned} \beta_{\mathrm{fBm}} &= 2H_{\mathrm{fBm}} + 1 \\ \beta_{\mathrm{fGn}} &= 2H_{\mathrm{fGn}} + 1 \cong 2(H_{\mathrm{fBm}} - 1) + 1 = 2H_{\mathrm{fBm}} - 1. \end{aligned} \quad (13)$$

While H varies from 1 to -1 from fBm to fGn, we realize from eqs (1)–(4) that Hu will always lie between 0 and 1 (Mandelbrot & Van Ness 1968; Mandelbrot & Wallis 1969). Thus, in general, H should not be considered as equivalent to Hu. Ignoring the difference between H and Hu can lead to confusing results. What has often been assumed is to make $Hu_{\mathrm{fBm}} = Hu_{\mathrm{fGn}} = H_{\mathrm{fBm}}$, and because of this, eq. (3) is often unjustifiably simplified to

$$\begin{aligned} \beta_{\mathrm{fBm}} &= 2Hu + 1 \\ \beta_{\mathrm{fGn}} &= 2Hu - 1. \end{aligned} \quad (14)$$

In reality, $Hu_{\mathrm{fGn}} \approx H_{\mathrm{fBm}}$ but $Hu_{\mathrm{fBm}} \neq Hu_{\mathrm{fGn}}$ and $Hu_{\mathrm{fBm}} \neq H_{\mathrm{fBm}}$. Eq. (14) is to keep Hu unchanged from $1 \le \beta \le 3$ to $-1 \le \beta \le 1$ and is actually a sacrifice, realizing that the meaning of Hu has been altered from $1 \le \beta \le 3$ to $-1 \le \beta \le 1$, because theoretically the relation $\beta = 2H + 1$ holds to be true for $-\infty < \beta < +\infty$. In order for eq. (14) to be meaningful, one has to assume that Hu of fBm is Hu from fGn, the successive increments of fBm. Without this common consensus, results of R/S analysis from different authors are incompatible and significant differences exist between H and Hu. To avoid these problems we have to clearly identify whether the time-series under study is analogous to fBm or fGn. If it is analogous to a fBm with spectrum power $1 \le \beta \le 3$, we have to find the subsequent increments of this time-series and then apply R/S to the new incremental series. Otherwise, as will be shown later in the next section, H will not equal Hu and Hu is almost certainly larger than 0.85, making Hu a very limited quantity to be used in characterizing fBm-like time-series. This is equivalent to saying that, in a strict sense, Hu is only better suited for capturing the properties of noise, such as fGn with Hu between 0 and 1, but not for random walks such as fBm.

Because of the delicate relationship between Hu and H, extra caution needs to be taken in relating Hu and H to the fractal dimension (D). For self-affine time-series, Voss (1985) derived the following approximate relationship between the spectrum power β and the fractal dimension D for fBm ($1 \le \beta \le 3$):

$$\beta = 5 - 2D. \quad (15)$$

From eq. (7), we have the much celebrated relationship

$$D = 2 - H. \quad (16)$$

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A direct analytical relationship between D and Hu has not been observed. However, for fGn, because $Hu_{\rm fGn} \approx H_{\rm fBm}$, eq. (16) can be extended to Hu so we can have

$$D \approx 2 - Hu. \tag{17}$$

This relationship cannot be extended for fBm because $Hu_{\rm fBm} \neq H_{\rm fBm}$. In order for eq. (17) to be applicable for fBm, we first need to obtain the associated fGn from fBm, calculate Hu from fGn, and take it as Hu of fBm.

We wish to point out that practically there is nothing wrong with applying R/S to fBm directly, but the Hu so obtained is not comparable to H from the power spectrum. In geophysical applications such as reservoir simulations, we value H as a better quantity than Hu because H captures the complexities of the data more accurately, while Hu is confined between 0 and 1. It is more realistic to say that R/S approximates the true scaling exponent H in eq. (5), but care must be taken when we relate H to Hu because these two quantities are not always interchangeable. The R/S technique can be used as a secondary method to roughly estimate H, mainly to validate the results from the spectrum method.

3 NUMERICAL SIMULATIONS

Using a frequency domain algorithm, we simulate 21 fBm series, each with 4096 data points. The input H for fBm ranges from 0 to 1 with a constant interval of 0.05. We could also simulate 21 fGn series using the same algorithm, but for the purpose of this article, we obtain the 21 fGn series directly from the 21 fBm series by taking successive increments. We then apply the R/S method to both fGn and fBm series to estimate Hu. It is expected that the H value from the power spectrum should be very close to the input H, for the simulation is itself based on a spectrum algorithm. It is nevertheless meaningful for us to check the robustness of our power spectrum method by estimating H on simulated fBm with the spectrum method. We also apply a third independent method called the variance of increment method to estimate H and validate our simulation.

Table 1 shows the results of analysis on simulated data. A rescaled-range analysis is applied on both fGn and fBm. We can immediately realize that R/S analysis on fBm almost surely gives an estimate of $Hu > 0.85$, as predicted. On the other hand R/S-estimated Hu from fGn is close to the estimate from the power spectrum and the variance of the increment analysis on fBm (Fig. 1). This further proves our earlier statement that, in order to make the estimate of Hu from R/S comparable to H from the power spectrum, we should apply R/S to the corresponding fGn series of fBm rather than to fBm itself. The final two columns of Table 1 show the spectrum power β of fBm and fGn. We can see that the relationships between β and Hu for fBm, $\beta_{\rm fBm} \approx 2Hu + 1$, and for fGn, $\beta_{\rm fGn} \approx 2Hu - 1$ are approximately true for our simulations, when the correct meaning of Hu is assigned.

Fig. 1 shows clearly that the rescaled range tends to overestimate the lower H values (for $H \lesssim 0.7$) and underestimate high H values (for $H \gtrsim 0.7$). This agrees well with results from other studies (Mandelbrot & Wallis 1969; Bassingthwaighte & Raymond 1994). Fig. 1 also shows an obvious linear correlation between Hu and H. An empirical relationship between Hu and H for fBm/fGn based on the least-squaresx regression is

$$H = 1.23Hu - 0.16. \tag{18}$$

Eq. (18) provides a simple way to roughly estimate the true scaling exponent H from a rescaled-range analysis for fGn/fBm. It is now straightforward to determine the crossover point where $H = Hu$. We find that when $H = Hu \approx 0.7$, the two methods should be able to give nearly identical results.

The initial 'transient zone' for the R/S technique persists when the input H is less than or equal to 0.5 (Fig. 2). Visually, the length of the transient zone is increasing from a lag spacing of 31 data points to approximately 63 data points when H increases from 0 to 0.5. The gradient of this zone is always higher than that of the rest of the data.

We find that when input H becomes larger than 0.5, the transient zone can no longer be clearly discernible from the rest, and it is safe to include this zone to calculate Hu. Meanwhile, however, another

Table 1. Results from R/S and power spectrum analyses on simulated fBm/fGn.

fBm no	Input H	R/S on fGn (Hu)	R/S on fBm (Hu)	Power spectrum (H)	Variance of increment (H)	β of fBm	β of fGn
1	0.0	0.173	0.900	0.012	0.054	1.024	−0.783
2	0.05	0.155	0.890	0.043	0.034	1.087	−0.763
3	0.1	0.226	0.938	0.114	0.139	1.228	−0.572
4	0.15	0.261	0.952	0.186	0.186	1.372	−0.452
5	0.2	0.309	0.947	0.206	0.219	1.411	−0.397
6	0.25	0.285	0.939	0.248	0.164	1.497	−0.327
7	0.3	0.323	0.970	0.324	0.272	1.648	−0.171
8	0.35	0.365	0.965	0.370	0.373	1.740	−0.104
9	0.4	0.443	0.995	0.411	0.371	1.822	0.000
10	0.45	0.508	0.996	0.463	0.529	1.927	0.117
11	0.5	0.569	1.004	0.504	0.531	2.009	0.197
12	0.55	0.580	0.991	0.541	0.561	2.081	0.261
13	0.6	0.676	0.999	0.631	0.576	2.261	0.407
14	0.65	0.670	0.993	0.654	0.575	2.308	0.485
15	0.7	0.764	0.999	0.700	0.756	2.399	0.579
16	0.75	0.786	0.997	0.772	0.702	2.543	0.714
17	0.8	0.784	1.000	0.807	0.688	2.614	0.763
18	0.85	0.789	0.966	0.849	0.637	2.699	0.855
19	0.9	0.837	1.005	0.922	0.714	2.844	1.003
20	0.95	0.863	1.000	0.986	0.618	2.971	1.097
21	1.0	0.886	1.000	1.012	0.542	3.024	1.188

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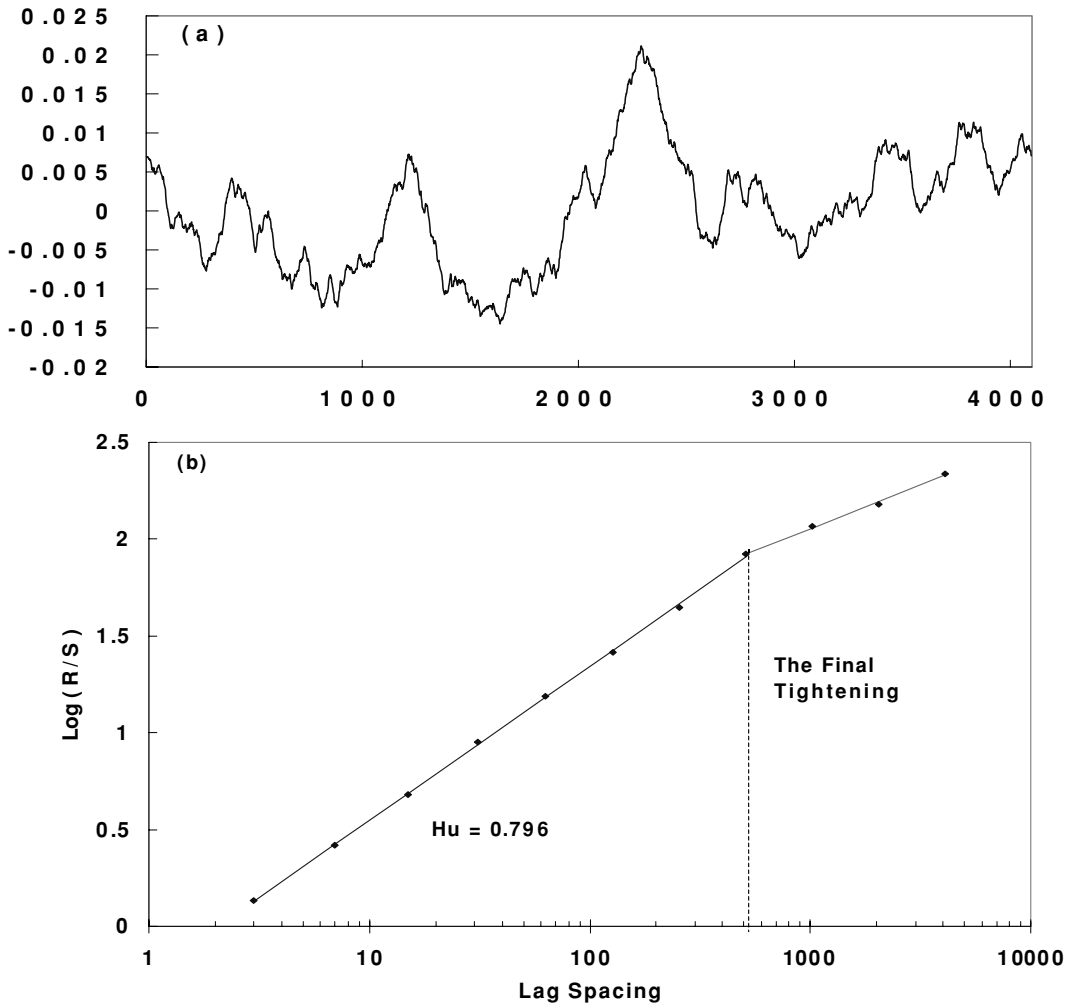


Figure 3. Plots showing the final tightening in the *R*/*S* analysis. (a) Simulated fBm no 18 with input $H = 0.85$. (b) *R*/*S* analysis on the simulation.

4.1 Gamma-ray data

The data, sampled at a constant rate of 0.5 ft for a total interval of 1984 ft, is from a well located in Panoma gas field, Grant County, southwest Kansas. Formation rocks penetrated by this well include shale, sandstone, siltstone, gypsum, anhydrite and dolomite. Fig. 4 shows the raw gamma-ray data (a), normalized data (b), data after moving average trend removal (c) and successive increments (d) from the raw data. To the right of these data series in Fig. 4 are their respective histograms. The raw data does not have a normal distribution, but all the others in the panel have distributions close to normal distributions. Unlike the raw gamma-ray data, we observe that modified data sets can be considered to be Gaussian/normal and stationary. Note here that normalization is not simply to extract the mean from the data and then divide the data by its standard deviation, because this will have little effect on removing short-wavelength variability. In fact, we estimate the time-varying variance and find the trend of the variance of non-stationary time-series data. We then normalize the data using the estimated variance.

Fig. 5 shows the rescaled range analysis on all data sets in Fig. 4. First, we see no transient zone for raw gamma-ray data, but distinct transient zones of various lengths (n) exist for modified data sets (b, c and d). Incremental series (d) gives the shortest transient zone. All observable transient zones are overlapping with the *R*/*S* plot of the raw data; in other words, transient zones from different data sets (b, c and d) have the same gradient on the *R*/*S* plot. For the raw gamma-ray data (Fig. 6a), the blindness of *R*/*S* to the transient zone may have an adverse effect on the *Hu* estimation if *R*/*S* is applied directly to raw data. This may also help to explain why various modifications to the raw data are necessary. Secondly, *R*/*S* analyses on modified data sets (b, c and d) give similar estimates of *Hu* regardless of how the raw data are modified. In this example, $Hu_{\text{increment}}$, $Hu_{\text{normalized}}$ and Hu_{MA} are 0.518, 0.378 and 0.54, respectively. On the other hand, *Hu* estimated from raw data is 0.91, which is substantially higher. Finally, we find that the incremental data (d) gives the best estimate of *Hu* because it has the shortest transient zone and the highest correlation coefficient ($R = 0.998$) from least-squares regression. In contrast, data with normalization (b) gives the lowest correlation coefficient of 0.955 and data with moving average trend removal (c) has the fewest data points for regression and has a correlation coefficient of 0.993. This leads to our conclusion that, while all the modified data give close estimates of *Hu*, the incremental series from the raw data gives the best estimate of *Hu*.

Unlike simulated fGn/fBm, well data suffer from the moving average effect of the logging tools (Shiomi *et al.* 1997). This effect will induce an abrupt drop of power at large wavenumbers (or frequencies) on the power spectrum plot (Fig. 6). The location of this breaking point is dependent on the resolution of the logging tools.

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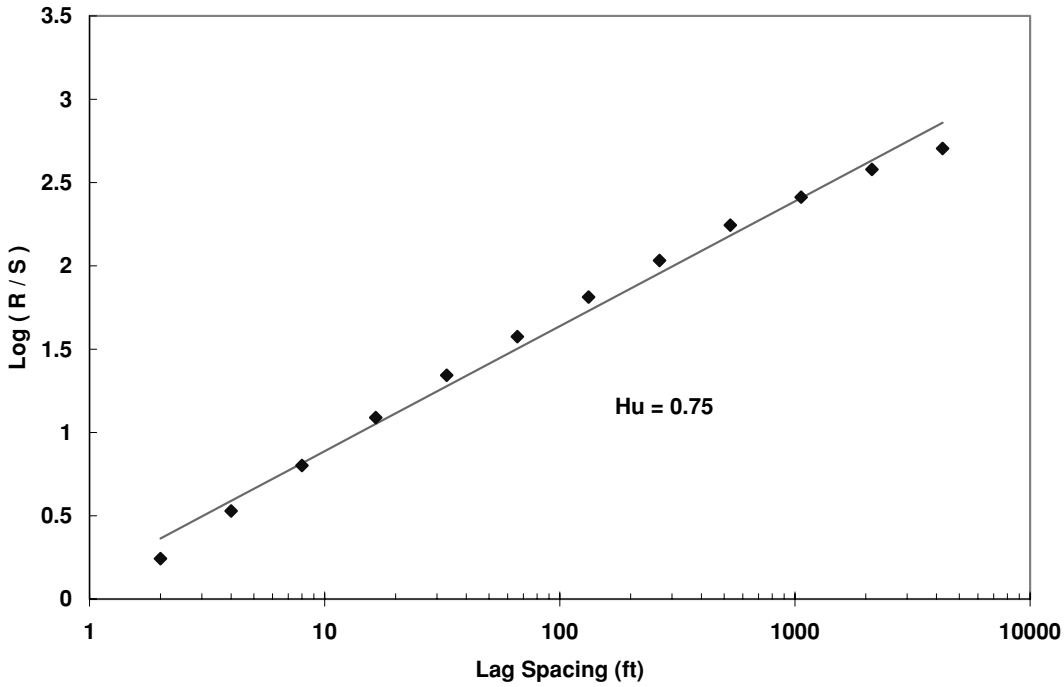


Figure 10. An example showing that, although a good linear regression ($R = 0.99$) can be made without identifying the existence of a transient zone, the estimated *Hu* is too high compared with H.

Table 2. *Hu*, H and D calculated from two well-logging data. (D is the fractal dimension.)

		Raw	Normalized	MA	Increment	Spectrum (H)
GR	*Hu*	0.91	0.38	0.54	0.52	0.27
	$D = 2 - Hu$ (or H)	1.09	1.62	1.46	1.48	1.73
	Box counting D_b	1.37				
	Regularization D_r	1.65				
	Transient zone lengths (ft)	0	70	94.6	9.3	
Sonic	*Hu*	0.89	0.51	0.3	0.25	0.017
	$D = 2 - Hu$ (or H)	1.11	1.49	1.7	1.75	1.983
	Box counting D_b	1.60				
	Regularization D_r	1.94				
	Transient zone lengths (ft)	0	212.7	170.3	11.2	

between $D_b = 1.60$ and $D_r = 1.94$. However, using $D = 2 - Hu$ based on the raw data, the estimated D is only 1.11, much smaller than D_b and D_r. D_b and D_r are also reasonably close to the fractal dimension D (=1.983) from $D = 2 - H$. This once again demonstrates that R/S analysis directly on the raw data is not appropriate for determining the fractal dimension.

5 CONCLUSIONS AND SUGGESTIONS

In this paper, we examined the paradoxical results often found in the literature concerning the Hurst exponent estimation, and provided an acceptable framework for more accurate analyses. Our arguments were made based on theoretical considerations, analyses from simulations of a group of fGn/fBm and studies on real data examples. We gained a better understanding of the nature of the well log data and argued that raw well-logging data should be considered as fBm-type time-series, which have a spectrum power $\beta \geq 1$. Consequently, for fBm-like data, in order to make the results from rescaled-range and power spectrum techniques consistent and comparable, and to make meaningful fractal dimension estimates, we need to use their incremental series, rather than the raw time-series themselves for R/S analysis. R/S analyses applied directly on the raw well data always tend to give erroneously high estimates of *Hu* (>0.85), making the technique itself less useful. On the other hand, R/S analyses on the incremental series give estimates of *Hu* close to H from power spectrum analyses on the raw data.

Other types of data modification, such as normalization or trend removal, could also be used to improve the analysis, and these modified new data series including the incremental series can give similar estimates of *Hu*. However, the increments have the shortest transient zones that make R/S analyses more accurate. It is also more objective to use the incremental series, because for each raw data series there exists only one unique incremental series that is also very easy to obtain. Data from normalization or trend removal could vary a lot depending on the specific data processing procedures. The degree of accuracy can also be viewed on the power spectrum plot where the increments have the broadest wavenumber/frequency ranges for linear regression.

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Since these modifications to raw well data introduce transient zones in the *R*/*S* analysis, it is very important to identify this zone and exclude it from calculating *Hu*; otherwise it will also result in erroneously high estimates of *Hu*. Comparing with the power spectrum analysis, we find that the transient zone lengths are closely connected to the regression lengths at long-wavelength region in the power spectrum. This reveals the close tie between these two independent data analysis methods.

Fractal models have been used for reservoir simulation (Hewett 1986; Hewett & Behrens 1990; Goggin *et al.* 1992). The decision in choosing the correct Hu/H is very important. For example, a realization based on $Hu = 0.11$ (Holliger 1996) will be quite different from a realization using $Hu = 0.7$ (Leary 1991), even for the same sonic data in Cajon Pass, CA. While we argue that raw well-logging data are of fBm type, in practice, for a new unknown data series it is wise to use the power spectrum method first to determine whether the spectrum power β is larger or smaller than 1. If $\beta \geq 1$, the data need to be considered as fBm-like and we use its increments for *R*/*S* analysis. There might be cases where the calculated $\beta < 1$ even for fBm-type data. This is most probably caused by an insufficient number of data points in the calculation. Both *R*/*S* and spectrum methods require a sufficient number of registered points for an accurate estimation. At least two independent methods are necessary for an accurate scaling analysis because they can provide a double-checking mechanism.

Finally, we realize that the Hurst exponent (*Hu*) should not be confused with the global scaling exponent (*H*) in practice. While these two quantities have some connections, they are fundamentally different and can only be considered equivalent under some constraints. The many discrepancies and conflicts in reported studies are a result of a lack of genuine understanding of the nature of both the data and the analysing techniques.

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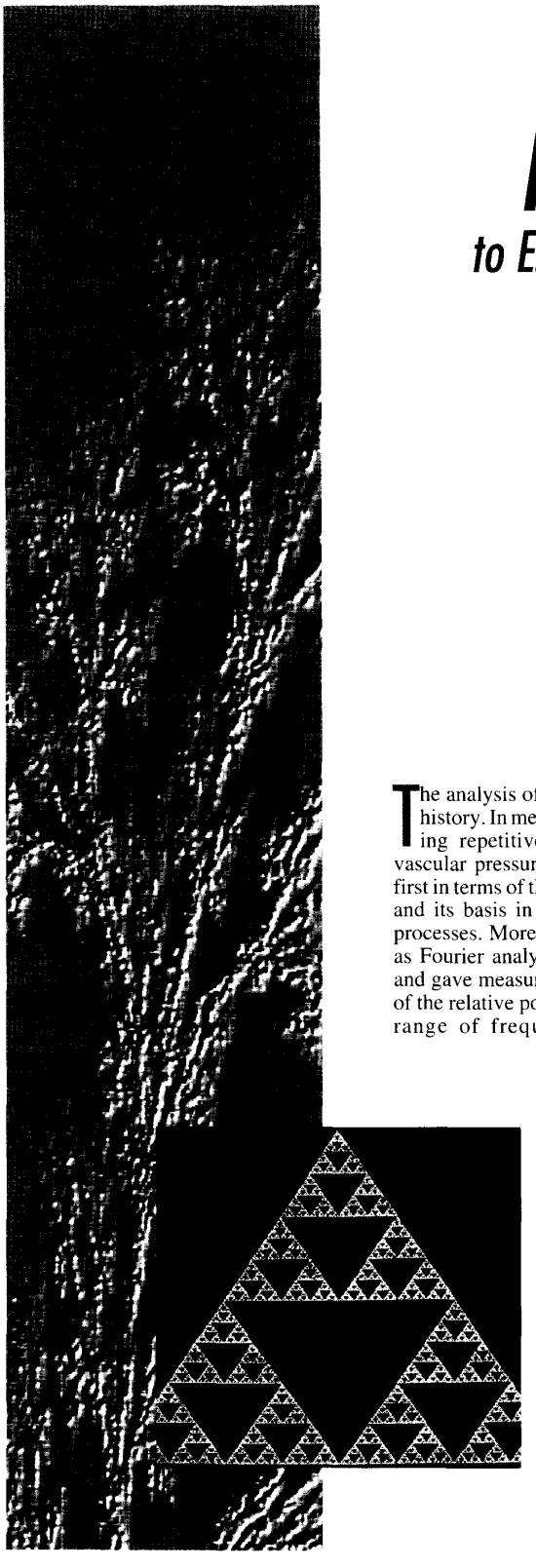
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Four Methods
to Estimate the Fractal Dimension from Self-Affine Signals

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The analysis of a time series has a deep history. In medicine, waveforms showing repetitive patterns (ECG, EEG, vascular pressure pulses) were analyzed first in terms of the specifics of the pattern and its basis in the underlying physical processes. More general techniques such as Fourier analysis were also applicable and gave measures of the periodicity and of the relative power contributions over a range of frequencies. Such analyses showed that the periodicities were not exact. Exact periodicity of cardiovascular events is an abnormal phenomenon [1], presumably because biological control systems are multidimensional, low-gain, and remarkably redundant. They balance well without any set-points. Nonlinearity in biological control systems is evident and leads, in a natural way, to their examination as low-dimension, chaotic dynamical systems [2], wherein there is a fractal attractor. Such systems operate within bounds and without apparent predictability.

Demonstrating the fractal or chaotic nature of a signal is valuable, for it provokes the development of new ways for discovering how the system works. It is in this spirit that we undertake to examine time series using one-dimensional approaches. Fractal systems (in time or space) have neighbor-to-neighbor correlation at all levels of scale. In spatial statistics, the variance of a local measure depends on the resolution of the measurement [3]. For instance, the variance of regional blood flow in the heart increases as the resolution of measurement is increased. In an experiment where the heart is divided into many sections for the measurement of the concentration of a flow-proportional marker (e.g., radioactive microspheres) in each of the sections [4], the relation between the resolution mass of observed tissue element, m, and the variance (Var) of regional blood flows is well described by a power law function:

$$Var(m) = Var(m_o)\left(\frac{m}{m_o}\right)^{2H-2} \quad (1)$$

where H is the Hurst coefficient, H = 2-D, and D is the fractal dimension lying between 1.0 and 2.0. The spatial distribution is considered to have fractal properties because it shows statistical

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self-similarity. Voss [5] showed that in a fractal system all of the moments of the distribution have the same power law slope. In our early analyses [6], we examined the relationship between the measure of variation (the coefficient of variation) and the resolution of the measurement. This was a simple one-dimensional spatial analysis which we label as RD analysis, where RD is the relative dispersion, i.e., the standard deviation divided by the mean.

Relative dispersion, correlation, rescaled range, and Fourier analyses all yield the fractal dimension, D

The dependence of the variance of a variable on the resolution was also found by the same fractal analysis of a time series [6], for example, the velocity of blood cells passing through a small artery as a function of time. If a signal, y(t), is divided into intervals of length, Δt, and the mean is calculated for each interval, one can also calculate the variance of these means; there will be a lower variance when the interval length is increased. The relationship between the variance and the sample time may fit a power law, and if so, the fractal dimension of the time series can be calculated by the same RD analysis as was used for a spatial measure.

In this article, we shall examine four methods for analyzing a one-dimensional time signal, y(t). The first of these is our RD analysis. The others are correlation analysis, rescaled range analysis, and power spectral analysis. All methods yield the fractal dimension, D.

When neighboring elements in a time series or a spatial distribution are positively correlated, the measured variance will drop less rapidly as the resolution is decreased. Based on this phenoenon, Bassingthwaighte and Beyer [7] reinvented the expression for the autocorrelogram (originally provided by Mandelbrot and van Ness [8]) from the relationship for covariances among neighboring units fulfilling Eq. 1. The one parameter of this function is the fractal dimension, and it can be estimated directly from an autocorrelogram (correlation analysis).

Another statistical measure from which the fractal dimension can be calculated is Hurst's rescaled range analysis [9, 10] for analyzing flows in rivers. The "range" is of the integral of deviations from the mean over an interval, and it is normalized by dividing by the standard deviation of the differences from the mean. Mandelbrot and Wallis published a series of papers was published on this analysis [11, 12, 13, 14, 15], showing that Hurst's method could be used to estimate the fractal dimension from a fractional Brownian noise. They modified the method by using the differences from a trend line, rather than differences from the mean.

The signals analyzed here are examples of fractional Brownian noises; they are characterized by a specific power spectrum of the form $S(f) \sim f^{-\beta}$, where f is frequency. Thus Fourier analysis [5, 16] is the fourth procedure used to estimate the fractal dimension. The 1/f noise, as it is frequently called, is common in nature, although the physical reason is not well understood. Recently, Bak et al. [18] proposed a general model for systems with a very high number of degrees of freedom, called 'self-organized criticality,' which accounts for both the occurrence of 1/f noise and fractals. Fractional Brownian noises and spatial fractals are, respectively, the temporal and the spatial fingerprints of a system that has evolved towards a critical state.

Do these four procedures to estimate the fractal dimension provide the same answer? Are they equally accurate? To find out, we generated signals with a known fractal dimension (D = 1.1 - 1.9), then used the four methods to estimate its value. The first problem is how to distinguish between the effects from the different signal generating algorithms, and the estimation methods. However, the estimation methods also perform better on long signals. The production of a truly fractal signal is itself a non-trivial exercise and may be as difficult as producing real random noise. So we must worry about the adequacy of the generating algorithms as well as the analysis algorithms. Some estimation procedures may require a very long signal, whereas others provide good results at shorter signal length. These four methods, however, are not exhaustive. The article by Fortin et al. in this issue [17] describes the use of a maximum likelihood estimator, another useful method.

The significance of D is that it describes, very compactly, the relation between the variance of the signal, and the time scale. No assumptions on the underlying system from which the signal was observed are made, and it is therefore a purely statistical tool. The fractal dimension estimated by these techniques from a time series is between one and two, i.e., for a one dimension function y(t), the fractal D lies between the Euclidean dimension E (= 1) and E + 1. (This is not at all the same fractal dimension that one attempts to derive from an embedding or correlation analysis [2], where one seeks a measure related to the order of the set of underlying differential equations.) For our analysis, $1 < D < 2$; values of D near 1 indicate high correlation or "memory" over time. (This can be seen, for example, in the nearest neighbor correlation coefficient $r_1 = 2^{3 - 2D} - 1$.) For random signals, D = 1.5 (Table 1).

We will test methods for estimating the fractal dimension of self-affine and self

Table I

D and H for Fractal Times Series

Correlation	D	H = 2 - D
Positive	1 < D < 1.5	1 > H > 0.5
Random	1.5	0.5
Negative	1.5 < D < 2	0.5 > H > 0

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from a fractal signal, β is reduced by two. Thus, for fractional Brownian noise, fBn, β is expected to be:

$$\beta = 2H - 1 \qquad (10)$$

Here again, a straight line is fitted from a log-log plot, and H is calculated from the slope β, as in the examples in Fig. 4. The influence for longer signals in the Fourier analysis is not clear from the individual plots, but the estimates are quite accurate even for shorter signals.

Rescaled Range Analysis

Edwin Hurst [9, 10] a hydrology engineer, had a statistical problem when designing the Aswan dam on the Nile River. He wanted to know how high the dam should be to contain all incoming river water and provide a constant outflow. When the inflow of the dam is a stationary random variable, the water level of the lake is the integral of the differences of the inflow and the constant outflow. He took data on the annual flow in the Nile and other rivers, and computed the range, R(u), defined as the maximum value minus the minimum value of the integral of the differences of the annual flow from the mean flow, divided by the standard deviation of the yearly inflow for different time spans ("time lag" u), averaged at different starting times, t_0. He found that a power law relation between the quotient of the range $R(t_0, u)$ and the standard deviation, S(u), of the inflow and the time span u described the data very well:

$$\frac{R(u)}{S(u)} \approx u^H \qquad (11)$$

where R/S is called the rescaled range. The longer time a dam should be able to hold all inflow, the higher it must be.

Mandelbrot and Wallis used Hurst's rescaled range analysis as an estimation procedure to estimate the fractal dimension of a signal [8, 11, 12, 13, 14, 15]. The quantity $R(t_0, u)/S(t_0, u)$ is calculated for subsets of the data over intervals of time lag u, starting from each of many starting points, t_0 separated by some interval, $\Delta t > u$, so that there are no overlapping subsets. The range is:

$$R(t_o, u) = MAX\{X(t_o, u)\} - MIN\{X(t_o, u)\} \qquad (12)$$

where:

$$X(t_o, u) = \int_{t_o}^{t_o+u} [B(s) - \langle B(s)\rangle]\, ds \qquad (13)$$

and <B(s)> denotes the mean value of the fBn over the interval $B(t_0, t_0+u)$. The standard deviation $S(t_0, u)$ is the standard deviation of B(t) over the same interval. For a data set of N evenly spaced observations, there are $N/(u/\Delta t)$ estimates of R/S obtainable from non-overlapping intervals of length u. We abbreviate $R(t_0, u)/S(t_0, u)$ to R(u)/S(u), which is a function of t_0 and u, and plot the individual calculations for each interval length, u, in Fig. 5, giving the "pox plot" [11]. A straight line is fitted in the log-log plot:

$$\log[R(u)/S(u)] = c + H \log(u) \qquad (14)$$

The individual points for a given u in Fig. 5 are calculated independently of each other, since there is no overlap between subsets. Using the GGOPT optimizer or a maximum likelihood estimator as used by Fortin et al. [17] gives a minimally biased estimate of H, which is better than fitting the means of R/S at each u.

The estimations from the R/S analysis depend on the way the subsets from the signal are taken to calculate the rescaled range. With longer signals, larger lags can be used, along with more points per lag. Also, the length of the shortest lag can be increased, which improves the fit further, as reported by Mandelbrot and Wallis [11-15].

Correlation and rescaled range analyses yield biased results under many circumstances

Results

In Fig. 6, the results are summarized for estimates obtained with the four methods for ten signals generated by the spectral synthesis method (SSM) and the successive random addition method (SRA) for H = 0.75. Several conclusions can be drawn from this Figure. Except from the R/S analysis and the Fourier analysis, the estimates of H tend to be too low. Some of the deviation may be blamed on the synthesis algorithm. For instance, deviations from the preset value of H = 0.75 with signals generated by the SSM algorithm are smaller than the deviations with the SRA algorithm (Figs 6a, 6b).

In general, estimates from the Fourier analysis come closest to the correct value of H = 0.75, and have a small standard deviation. The estimates from the RD and correlation analyses are low. The correlation estimates may be biased towards 0.5 because correlation coefficients over too

Table 2

Estimates of H from y(*t*), ten trials each; actual H = 0.75

Synthesis Algorithm	Analysis Method	Estimates of H at different N (± SD)		
		N = 512	N = 8192	N = 32768
SSM	RD	0.652 (0.076)	0.735 (0.028)	0.7043 (0.027)
	Cor	0.678 (0.036)	0.736 (0.018)	0.724 (0.007)
	R/S	0.736 (0.169)	0.778 (0.042)	0.738 (0.032)
	Spectrum	0.764 (0.042)	0.773 (0.013)	0.770 (0.008)
SRA	RD	0.639 (0.065)	0.698 (0.027)	0.716 (0.033)
	Cor	0.622 (0.057)	0.676 (0.023)	0.696 (0.015)
	R/S	0.727 (0.054)	0.740 (0.046)	0.748 (0.047)
	Spectrum	0.748 (0.033)	0.738 (0.018)	0.742 (0.003)

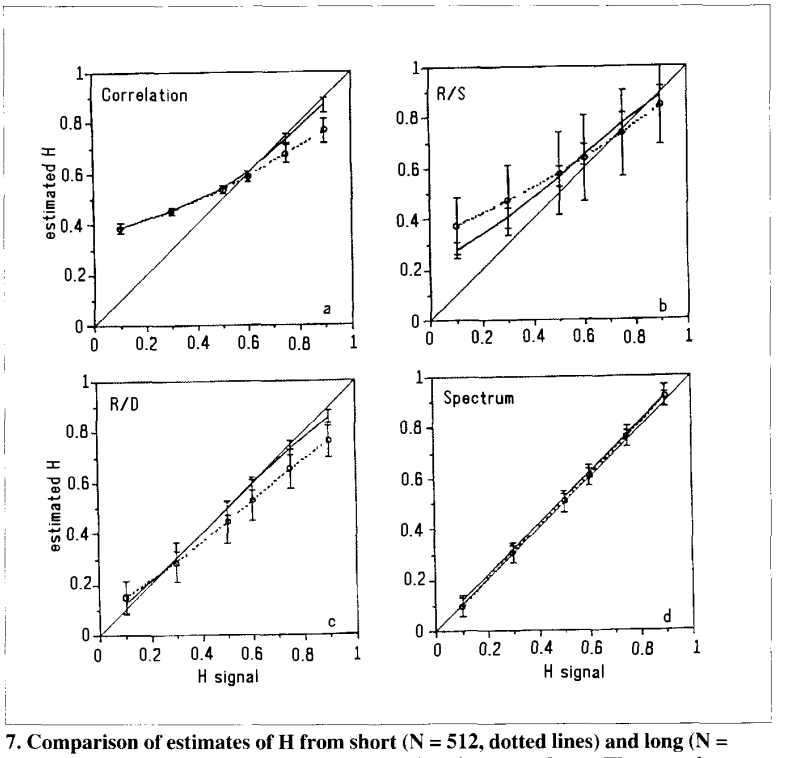
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large distances are taken into account. The mean of the estimations from the rescaled range analysis is fairly good, but the standard deviation is very large. The estimates from Fig. 6 are listed in Table 2.

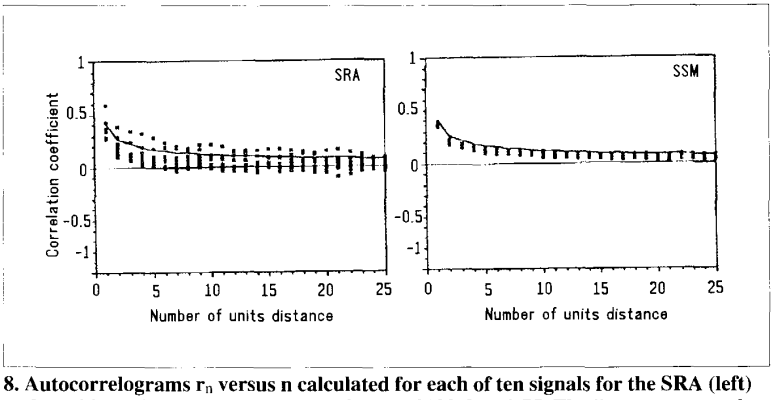
The estimates of H obtained via these procedures are presented in Fig. 7 for signals with true values of H from 0.1 to 0.9. The Fourier analysis yields the best results (right lower panel). It is apparent that the correlation analysis yields estimates biased towards H = 0.5. The correlation estimates for high values of H are clearly improved when a longer signal is analyzed.

Estimates of the R/S analysis tend to be too high, but with a large standard deviation (Fig. 7, right upper panel). The RD analysis yields excellent results with 8192 points. Even with only 512 points, at H = 0.1 and 0.3, the RD analysis provides almost correct values, underestimating at higher H.

An important difference between the correlation structure of signals generated by the SRA and SSM algorithm is shown in Fig. 8. The first 25 correlation coefficients of 10 signals for each algorithm are plotted along with Eq. 7 with H = 0.75. The SRA algorithm appears to produce a more nearly randomly structured signal, as the estimates tend toward 0.5 and therefore fall mostly below the expected correlation for H = 0.75 (Fig. 8a). The values for the correlations are too low, and the scatter is large. The SSM produced signals whose autocorrelograms are only a little below the model function equation, Eq. 7 (Fig. 8b).



7. Comparison of estimates of H from short (N = 512, dotted lines) and long (N = 8192, solid lines) signals from the different estimation procedures. The error bars denote standard deviations of estimates of H from ten signals at each H. Signals were generated by the SSM algorithm. Correlation analysis (a); R/S analysis (b); RD analysis (c); spectral analysis (d).



8. Autocorrelograms r_n versus n calculated for each of ten signals for the SRA (left) and the SSM (right) algorithms, both for N = 8192, H = 0.75. The lines represent the expected correlation function with H = 0.75.

Discussion

A major problem of the work described above involves the question: "What is being tested, the estimation procedures by the signal generating algorithm, or the signal generating algorithm by the estimation procedures?" When a signal generating algorithm produces a signal with a different fractal dimension from the dimension corresponding to the parameter H, it cannot be expected that the estimation procedures will be unbiased with respect to the preset parameter value. But because the estimations were done on the same signals, it is still possible to determine which of the methods comes closest to the preset value, and the variances of the estimation procedures can be compared.

The variance of estimates from the R/S analysis is much larger than those of the other estimation procedures. This occurs because in the R/S analysis, the signal must be divided into several subintervals, which can be done in many ways. The results for the R/S analysis clearly depend on the number of subintervals (the number of points in Fig. 5 at each lag), the minimum and maximum lags. It is not clear how important the number of subintervals is. Mandelbrot and Wallis noted that the smallest lags (under 30) can't be expected to be on the line, and should not be included in the procedure. In this article, we chose not to use overlapping subintervals, so that no extra dependence between these subintervals occurs. Hurst reported that the estimates from the R/S analysis might be biased towards 0.72, in agreement with our finding at short signal lengths. However, at long signal lengths the bias seemed to be toward H = 0.8. It is not clear why this would occur. Further work needs to be done, especially on the influence of picking the particular subintervals from the main signal. As can be seen with the correlation analysis for signals of length 512, 8192 and 32768, in Fig. 3, the SSM

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algorithm produces a better correlation structure for longer signals. From Fig. 6, it can be seen that the same applies to the SRA algorithm, because the estimations also become progressively better; compare Figs. 6b, 6d, and 6f.

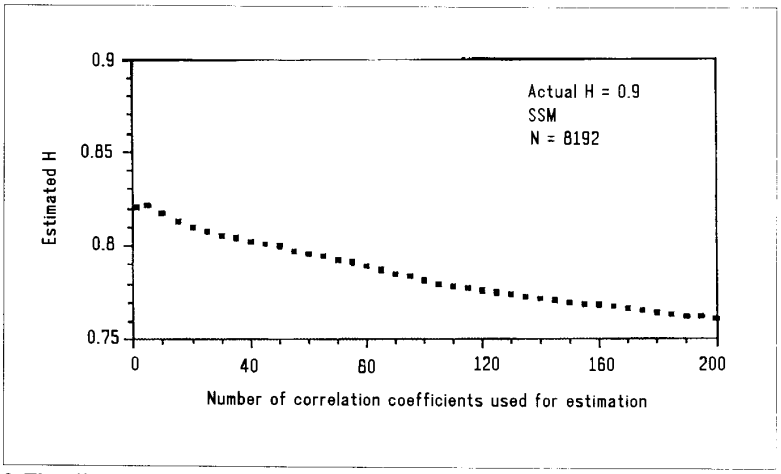
In the estimation of H from the correlogram, the estimates of r are biased towards zero when n is large. Figure 9 shows the estimations of H as a function of lags included in the correlogram for the fit of the correlation function, Eq. 6. The signal was generated with the SSM algorithm with an H = 0.9. The bias towards 0.5 becomes progressively more apparent when more distant correlation coefficients are included. With n large, the calculation of r_n is also rendered less accurate, simply because there are fewer pairs of values to correlate at large distances. These difficulties are exacerbated when H is closer to 0.5, and lessened when H is > 0.8.

Spectral analysis yields the least biased results, and the lowest variance in estimates of D

In the range of H between 0 and 0.5, all estimations except for the Fourier analysis and RD analysis with long signals are not very good. The reason is that the data points from which the Fourier analysis estimates the fractal dimension are independent of each other. With the correlogram this is not the case. The correlations are negative for the first few lags, but very quickly become scattered around zero. When too many correlation coefficients are included in the estimation procedure, a biased estimate towards H = 0.5 is obtained. The estimation from only the first autocorrelation coefficient is not good, either. The first autocorrelation coefficient as a function of H is described by Eq. 5 [23]. Why a first autocorrelation coefficient of less than -0.5 is not possible for a fractal signal is not well understood.

Of all the methods discussed, the Fourier method is generally best. However, the variance (or SD) of the estimate is still larger than that given by the maximum likelihood methods (MLE) in Lundahl et al. [16] and Fortin et al. [17]. For example, with the Fourier method and N = 512 points, the SD is 0.033. Use of the MLE yeilds an SD = 0.01. The major liability of the MLE is its extensive computation time.

The procedure to calculate the fractal dimension of the strange attractor of the underlying dynamical system [2] is not discussed here. The fractal dimension of the strange attractor is conceptually different: it gives a measure of the minimal order of a system, the order N of N-dimensional phase space in which the time signal and its derivatives (or the order of the set of differential equations defining the system) can be mapped, plus the degree to which the signal fills the space. Although one would think that a little noise would encourage overestimating the dimension, in the situation where the system is relatively stable, the noise actually has the effect of revealing the dimensionality of the attractor more fully [29]. A main drawback of the analysis in [2] is that it requires a very large (long) time series in order to converge on the estimate of the dimension.



9. The effect of the number of correlation coefficients that are included in the estimation for H from a signal generated by the SSM algorithm, with N = 8129 and H = 0.9.

Conclusion

The fractal dimension provides a measure of the heterogeneity of the system and takes the correlation structure into account: D ≠ 1.5 implies that we are not dealing with a random distribution, but that there is correlation between adjacent regions. We compared four numerical methods to estimate the fractal dimension. Correlation analysis and rescaled range analysis yield seriously biased results under many circumstances. Relative dispersion analysis is well suited for long signals. Spectral analysis gives the least biased results, and also has lowest variance in the estimates of the fractal dimension. The same methods can be used for the one-dimensional analysis of spatial signals, as Hurst et al. [10] did for the analysis of thicknesses of tree growth rings, and mud layers. For the analysis of two- and three-dimensional signals, the methods can be extended to account for anisotropy, making the analysis more complicated but adhering to the same basic theory.

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GEOPHYSICAL RESEARCH LETTERS, VOL. 16, NO. 7, PAGES 673-676, JULY 1989

ON THE USE OF SPECTRAL METHODS FOR THE DETERMINATION OF FRACTAL DIMENSION

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Abstract. This paper discusses the use of fractal theory for quantitative analysis of profiles from two-dimensional surfaces. The theoretical relationship between spectral parameters and fractal dimension is examined, including the conditions under which the derived relationship between spectral parameters and fractal dimension by Voss (1985) is valid, and the limitations in making inferences from spectral parameters. Applications of fractal theory to geophysical data are also discussed. In particular, it is shown that an amplitude spectrum with a decay corresponding to a fractal dimension of 1.5 can result from the concatenation of time series with decays corresponding to different fractal dimensions. It has been shown that the spectral density of a fractal distribution will be characterized by a power-law decay, but this paper illustrates that a power-law decay is not sufficient to identify a fractal distribution. Although this paper discusses applications to the study of Earth topography, the results are applicable to the study of any one-dimensional profile.

Introduction

Although plate-tectonics provides a comprehensive view of large-scale tectonic processes and geological structures, the formidable complexity of topography has previously hindered efforts to understand the statistical character of Earth topography in any generality. Since Mandelbrot (1967) first introduced the term fractal, many researchers in Earth sciences have looked to fractal theory to provide a framework for quantifying spatial characteristics of topography, faulting, etc. The well-known scale-invariance of many geological and geophysical phenomena suggest that it is appropriate to model observations with a fractal distribution (Turcotte, 1988). The usefulness of interpreting geophysical observations in terms of fractal distribution goes beyond a simple recharacterization of observations because it sets the stage for development of new techniques to understand the statistical nature of topography and other geophysical processes. While details of the geology of individual regions can be understood through conventional geological and geophysical methods, focusing on the spectral and statistical character of topography rather than the spatial details provides a context from which individual pieces can be viewed as part of a larger picture.

As strictly defined, a self-affine fractal distribution describes phenomena which are continuous but not differentiable and correlated over many length scales. A profile is self-similar if its statistical properties are not changed when the profile is decimated or when parts of the profile are enlarged. Mandelbrot (1983) introduced the term self-affine to describe distributions whose statistical properties are unchanged after enlargement only after the x and y axes have been rescaled by different proportions.

Spectral and Fractal Theory

Fractal analysis has been used to look at sea-floor and continental topography (e.gl Sayles and Thomas, 1978; Berry and Hannay, 1978; Bell, 1975; Fox and Hayes, 1985; Brown and Scholz, 1985; Malinverno, 1988; Gilbert, 1989; Gilbert and Malinverno, 1989; Goff and Jordon, 1989a). A method of relating the fractal dimension, D, to quantities that can be measured from discrete data is essential to applications of fractal theory. Voss (1985) relates D to properties of the power spectral density, S(f), the 2-point autocorrelation, G(τ), and to the mean square increments, $V_i(f)$ of a function. These are properties which can be estimated from data, although the strict definition of spectral density must be extended to non-stationary functions (Voss, 1985; Mandelbrot and Van Ness, 1968).

As discussed by Voss (1985), a function z(t) will have a Fourier transform for $0 < t < T$ given by

$$Z(f,T) = \frac{1}{T}\int_0^T z(t)e^{2\pi ift}dt \qquad (1)$$

and a spectral density of

$$S(f) = T|Z(f,T)|^2 \qquad T\rightarrow\infty \qquad (2)$$

The two-point autocorrelation is given by

$$G(\tau) = \langle z(t)z(t+\tau)\rangle - \langle z(t)\rangle^2 \qquad (3)$$

where $\langle\rangle$ denotes expected value. G(τ) is related to the mean-square increments by

$$M_i(\tau) = \langle |z(t+\tau)-z(t)|^2\rangle = 2[\langle z^2\rangle-\langle z\rangle^2] - 2G(\tau) \qquad (4)$$

In the case of a random, stationary process, G(τ) can be related to the spectral density by the Wiener-Khintchine relation,

$$G(\tau) = \int_0^\infty S(f)\cos(2\pi ft)df \qquad (5)$$

Because the mean-square increments of a fractal distribution can be related to the fractal dimension (Voss, 1985), equations (1)-(5) have been used to relate spectral parameters to fractal parameters. The Weiner-Kintchine relation must be extended to non-stationary signals (Voss, 1985; Mandelbrot and Van Ness, 1968). It will be argued later that this extention is not, in fact, valid. If one assumes a power spectrum with power-law form, $S(f) \alpha f^{-\beta}$, then

$$D = \frac{5-\beta}{2} \qquad (6)$$

for a two-dimensional profile (Berry and Lewis, 1980). As defined, estimates of D from equation (6) are not necessarily consistent with the fractal (Hausdorff) dimension. Because D for a profile must lie between 1 (completely differentiable) to 2 (so rough as to fill up all of 2-dimensional space), β is apparently constrained to be between 1 and 3. Although equation (6) has been applied with success to real data, the derivation of (6) given by Voss (1985) is incomplete. Most fundamentally, once the complex Fourier transform is reduced to the spectral density in equation (2), then all phase information is lost. Any results from subsequent derivations will be valid if and only if an independent verification of random phase is demonstrated. Even given random phase, the derivation is further flawed when the Wiener-Khinctine relationship is extended to apply to non-random processes. This paper will discuss the validity of equation (6) in light of these concerns, as well as the "paradox" of $\beta > 3$.

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Spectral Density and Autocorrelation

Voss (1985) states that that equation (5) can be used to obtain $G(\tau) \alpha \tau^{\beta-1}$. However, this direct calculation is only valid for $0 < \beta < 1$. In fact, equation (5) becomes infinite when β is larger than one because the singularity at the origin becomes stronger than a simple pole (e.g. Marsden, 1973). But any real data set is integrated from a discrete minimum frequency, f_{min}. Replacing the lower limit of integration and letting $x = 2\pi ft$, equation (5) becomes

$$G(\tau) = (2\pi\tau^{\beta-1})\int_{x_{min}}^{\infty} \frac{1}{x^{\beta}}\cos x dx = (2\pi\tau^{\beta-1})I(x) \qquad (7)$$

Thus $G(\tau)$ will be proportional to $\tau^{\beta-1}$ as long as $I(x)$ is finite and not dependent on τ. $I(x)$ can be integrated by parts n times to give equations of the following form

$$I(x) = \frac{1}{1-\beta}\left[\frac{\cos(x)}{x^{\beta-1}} + \frac{1}{2-\beta}\left[\frac{\sin x}{x^{\beta-2}} - \frac{1}{3-\beta}\left[\frac{\cos(x)}{x^{\beta-3}} - \int_{x_{min}}^{\infty}\frac{\sin(x)}{x^{\beta-3}}\right]\right]\right] \qquad (8)$$

until $\beta-n$ lies between 0 and 1. The integrated terms in equation (8) are evaluated from x_{min} to ∞. Clearly, these terms are zero at the upper limit. At the lower limit, $x_{min} = 2\pi\tau_{min}f_{min}$. If $z(t)$ is sampled at δt, f_{min} will be given by $1/(\delta t * n)$, where n is the number of points in the time series. Since τ_{min} is just given by δt, we have

$$x_{min} = 2\pi\left(\frac{1}{(n\delta t)}\right)(\delta t) = \frac{2\pi}{n} \qquad (9)$$

Thus the largest of the integrated terms goes as $n^{\beta-1}$, which is finite and independent of τ. The remaining integral in equation (8) will either be given by

$$\int_{x_{min}}^{\infty}\frac{\cos(x)}{x^{\beta-n}}dx \qquad 0 < \beta-n < 1 \qquad (10)$$

or by the same equation with sin(x) replacing cos(x). Because x_{min} is independent of τ and because

$$\int_{x_{min}}^{\infty}\frac{\cos(x)}{x^{\beta-n}}dx < \int_{0}^{\infty}\frac{\cos(x)}{x^{\beta-n}}dx = \Gamma(j)\cos\left[\frac{(j)\pi}{2}\right] \qquad (11)$$

(where $j=1+n-\beta$) then equation (10) is also finite and independent of τ_{min}. The same argument holds when cos(x) is replaced by sin(x). Thus the result $G(\tau) \alpha \tau^{\beta-1}$ can be extended for discrete time series to $1<\beta<3$. This same result for 1, 2, and 3-dimensional spectra is found by Andrews (1981), although his paper does not examine rigorously the conditions of convergence of equations of the form of equation (7).

The "Paradox" of $\beta > 3$

If $G(\tau) \alpha \tau^{\beta-1}$, then the mean square increments of z(t), given by equation (4), are also proportional to $\tau^{\beta-1}$. If $\beta > 3$, the mean-square increments are twice differentiable near the origin and the function, z(t), is itself differentiable (Journel and Huijbregts, 1978). Thus $\beta > 3$ does not present the mathematical paradox of $D < 1$ as implied by equation (5). Rather, $\beta > 3$ implies $D = 1$; that the function is differentiable. Although the large class of continuous, differentiable functions varies considerably in the degree of what is non-rigorously thought of as wiggliness, the fractal dimension, as defined, cannot distinguish between the character of any function in this class.

The breakdown of equation (4) is suggested by numerical results given by Malinverno (1989) and explained by Brown and Scholz (1985) by examining the second moment of the power spectrum . Malinverno (1989) compares the fractal dimension estimated from spectral slope with D determined from the behavior of the mean-square increments at short range, and finds that the two estimates

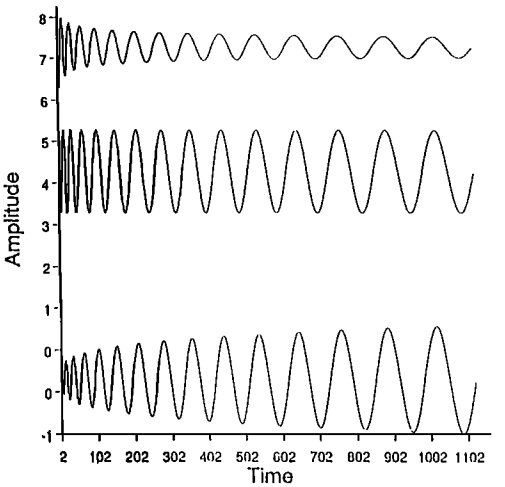


Fig. 1a. Time (or space) series generated by scaling sine waves with differing functions of time (or space). Three series are offset vertically for clarity.

are equal only for D=1.5; they are slightly different for all other values of D between 1 and 2 . This illustrates the previously-mentioned problem with extensions of the Wiener-Khinctine relation: Given random phase and D=1.5, the profile is a random walk, and equation (5) is valid. For other values of D, however, then even with random phase, then a profile or time series is not a random process and there is no justification for the use of equation (5). Numerical results of Malinverno (1989) and Fox (1989) suggest, however, that for $1<\beta<3$, equation (6) can be a reasonable approximation.

Application to Analysis of Data

The notion that a function with a specified amplitude spectrum will be critically dependent on its phase spectrum is by no means new, but the extensive utilization of equations (2)-(6) illustrates that this point has not been consistently appreciated. To present a few simple illustrations, one can construct time series with specified amplitude spectra and distinctly non-random phase. Figure 1 shows functions made up of sine waves with increasing period.

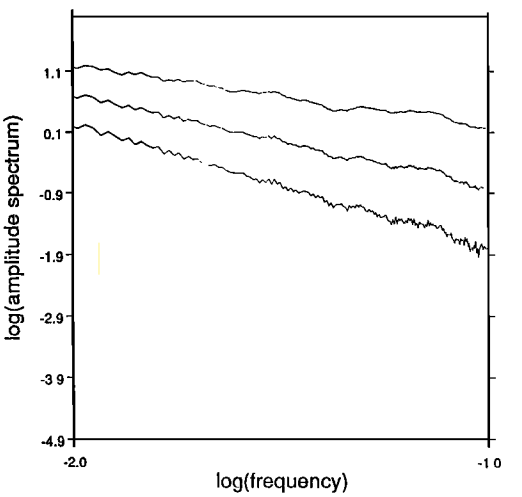


Fig. 1b. Amplitude spectra of the time series shown in figure 1a. The spectra are shown for frequencies corresponding to the frequency range of the sine waves in figure 1a, roughly 0.01 Hz to 0.1 Hz

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One can obtain power spectral amplitudes with $S(f) \alpha f^{-\beta}$ behavior over the frequency bands corresponding to the periods in the sine waves. Depending on how the amplitudes are scaled, β can take on a range of values. Topographic or bathymetric data sets are unlikely to look anything like the curves in figure 1, but processes such as folding could produce features with non-random phase. The potential danger of computing spectra from non-stationary profiles

was recognized by Fox and Hayes (1985) who showed that, while a spectrum of a long bathymetric profile crossing several "statistically homogeneous provinces" corresponds to D = 1.5 , spectral slopes taken from individual quasi-stationary profiles give a range of spectral slope that generally do not correspond to D of 1.5 (Figure 2). Numerical studies by Fox (1989) also suggest a general tendency of long non-stationary profiles to tend towards a "self-similar" model. Fox and Hayes (1985) make only passing reference to interpretation in terms of fractal distribution . It is clear from the behavior of the individual profiles that it would be inappropriate to interpret D from the long profile as implying self-affinity because in that case, any part of the long profile would have the same statistical properities as any other.

The results of Fox and Hayes (1985) and Fox (1989) suggest that a fractal dimension of 1.5 might, in fact, be a limiting value obtained by summing time series' with different fractal dimensions. In fact, if the time series shown in figure 1a are concatenated (figure 3a), the amplitude spectrum has a decay very close to $\beta = 2$, or D = 1.5 over the frequency range of the sine waves (figure 3b). Sayles and Thomas (1978) argue that a $1/f^2$ spectrum is the result of a central limit theorem. However, what they show is that this spectrum results from a fractal distribution. A proof that results such as those shown in figure 3b are the result of a limit theorem requires the consideration of a sum of adjacent time series, each of which has an amplitude given by $1/f^{\beta}$, where β ranges between 1 and 3 and in part of a specified distribution. If a time series, z(t) is created by concatenating n time series, $z_i(t)$, each of which has a spectrum given by $1/f^{\beta_i}$,then

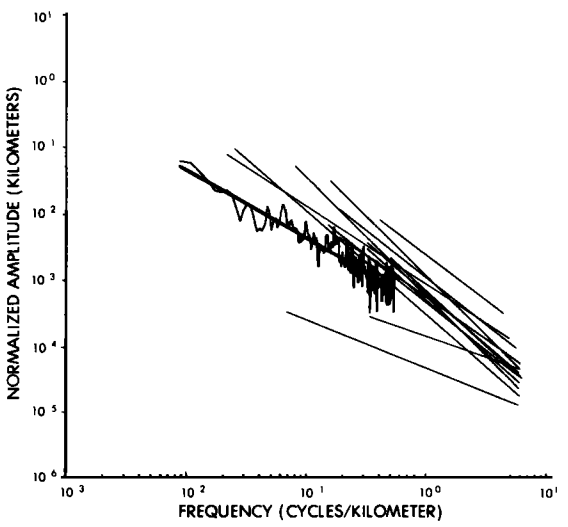


Fig. 2. Amplitude spectrum of long bathymetric profile extending almost 500 km from the Oregon coast. Superimposed line has slope corresponding to D of 1.5. Also shown are spectral slopes from individual segments which Fox and Hayes (1985) identify as representing quasi- stationary provinces on the seafloor. Reprinted from Fox and Hayes (1985).

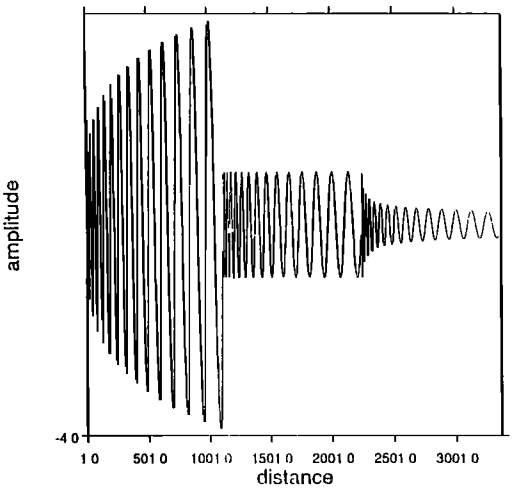


Figure 3a. Series generated by the concatenation of the three series' shown in figure 1a.

$$z(t) = \sum_{i=1}^{n}(H(t-t_i) - H(t-t_{i-1}))z_i(t) \qquad (12)$$

and the Fourier transform of z(t), S(f), is given by

$$S(f) = \int_{-\infty}^{\infty}\sum_{i=1}^{n}(H(t-t_i) - H(t-t_{i-1})z_i(t))e^{2\pi ift}\,dt \qquad (13)$$

where H(t) is the heavyside step function. Interchanging the order of integration and summation, and noting that the effect of the boxcar functions is to change the integration limits from $\pm\infty$ to t_{i-1},t_i. Ignoring windowing effects, one obtains,

$$S(f) \rightarrow \sum_{i=1}^{n}\frac{1}{f^{\beta_i}} \qquad (14)$$

If β_i are given by $m \pm \sigma_i$, then

$$S(f) = \frac{\sum_{i=1}^{n} f^{((n-1)m + \sum_j(1-\delta_{ij})\sigma_j)}}{f^{(nm + \sum_i^n \sigma_i)}} \qquad (15)$$

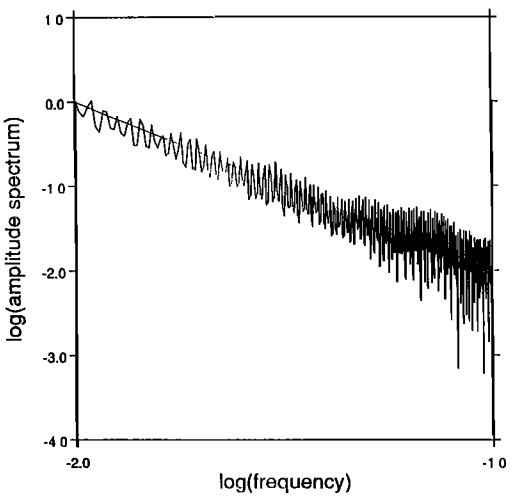


Figure 3b Amplitude spectrum of the concatenated time series shown in figure 3a. Straight line shows a $1/f^2$ fall-off between 0.01 Hz and 0.1 Hz.

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If the sum of σ_i averages to zero, then

$$S(f) \rightarrow \frac{nf^{(n-1)m}}{f^{nm}} = \frac{n}{f^m} \quad (16)$$

Thus a $1/f^2$ spectrum can result from a simple averaging process. It should be noted that the summed segments here are assumed to have consistent amplitudes. If one scales the amplitude of the individual segments differently, then the largest amplitude segments will dominate the spectrum of the concatenated time series.

A self-affine or self-similar fractal distribution is characterized by three essential characteristics: 1)It must have appropriate correlation over many length scales, 2)It must be continuous but not differentiable, and 3)It must have random phase. Criteria (1) can be tested by examining the behaviour of the amplitude spectrum, although Gilbert (1988) notes that care must be taken in making inferences from topographic data because data processing can affect spectral behaviour. The amplitude spectrum can also be used to determine whether or not a profile is differentiable, since a differentiable profile will have $\beta>3$. Testing for random phase can be accomplished by computing spectral parameters within a moving window within the time series, a "split moving window" method (Malinverno, 1988; Webster, 1973; Fox and Hayes, 1985; Goff and Jordan, 1989b). This type of test identifies boundaries between statistically homogeneous segments by computing spectra from a specified window length which is moved across the profile. Malinverno (1989) shows that statistically distinct segments can be identified by changes in spectra slope and intercept.

Conclusions

This paper has discussed the applicability of fractal theory to the analysis of geophysical, generally topographic, data. The theoretical relation between spectral parameters to fractal dimension (Voss, 1985) is discussed. While theoretical limitations are noted, it is argued that the results of Voss (1985) are good approximations under certain circumstances. The usefulness of the concepts of fractal distributions for the analysis of data, beyond recharacterization of observations, has been demonstrated in the analysis of topographic data. Fox and Hayes (1985), Malinverno (1989), and Goff and Jordan (1989b) present ways to identify distinct portions of the seafloor, and Malinverno and Gilbert (1989) interprets observed fractal dimensions in terms of a fractal faulting process. The elegant simplicity of fractal analysis is appealing, as is the recognition that universal physical principles appear to control the complexities of diverse types of physical and biological systems. What is elegant and attractive is not, however, necessarily strictly applicable. Studies of topography done by Brown and Scholz (1985), Gilbert (1989), and Goff and Jordan (1989b) suggest that spectra of topographic profiles do not obey a strict power-law decay. And, as discussed in this paper, the requirement of power-law decay of a spectrum is only a necessary, not a sufficient, condition for identifying a fractal distribution. Several researchers have combined data from different regions to form a composite spectrum with a $1/f^2$ decay and concluded that this demonstrates self-affinity of Earth topography (Sayles and Thomas, 1978; Turcotte, 1989). Results from this paper demonstrate that conclusions about Earth topography as a whole are premature, and that while statistically homogeneous provinces may be described by self-similar or self-affine distributions, analysis of topographic data will permit inferences about geological processes only if careful studies are made in limited geographical regions.

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Fractal Correction of Well Logging Curves*

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ABSTRACT

It is always significant for assessing and evaluation of oil-bearing layers, especially for well logging data processing and interpretation of non-marine oil beds to get more accurate physical properties in thin and inter-thin layers. This paper presents a definition of measures and the measure presents power law relation with the corresponded scale described by fractal theory. Thus, logging curves can be reconstructed according to this power law relation. This method uses the local structure nearby concurrent points to compensate the average effect of logging probes and measurement errors. As an example, deep and medium induced conductivity (IMPH and IDPH) curves in ODP Leg 127 Hole 797C are reconstructed or corrected. Corrected curves are with less adjacent effects through comparison of corrected curves with original one. And also, the power spectra of corrected well logging curve are abounding with more resolution components than the original one. Thus, fractal correction method makes the well logging more resoluble for thin beds.

KEY WORDS fractal, thin bed well logging, singularity of well logging, well logging correction.

INTRODUCTION

Because well logging measurements give an average results nearby concurrent depth points of the probes as shown in Fig. 1a, well logging curve does not supply the real geophysical properties at concurrent depth but a disturbed result of adjacent layers and borehole environment. How to retrieve real physical value has been a subject for well logging analysts and especially for Chinese well logging analysts who must mainly cope with non-marine sedimentary reservoir. Techniques for well logging data processing and interpretation in thin, inter-thin beds are a Chinese-characterized difficult subjects in oil exploration and development. Intrinsic methods to invest thin beds should work by probes design and calibration, though post-processing method such as deconvolution can also improve the resolution in some extent (Wang et al., 1997).

According to fractal theory, measurement is related with scale by power law (Pape et al., 2000; Cheng, 1999; Doveton, 1998; Van Wagoner, 1990). This is called scale invariant or self-affinity. So, by proper definition of well logging curve measure, power law could be certified and applied.

For an equally sampled well logging series (Fig. 1b), using an expanding scale window (Δh, $3\Delta h$, $5\Delta h$, ... Δh is acquisition interval) centered at concurrent points K (at depth $K \cdot \Delta h$) a series of covers can be built up. With these covers, measure is defined as

$$\mu(K,\varepsilon) = \sum_{i=K-\frac{M}{2}}^{K+\frac{M}{2}} x(i)\Delta h \tag{1}$$

where $\mu(K,\varepsilon)$ is measure at point K under cover scale $\varepsilon = M \cdot \Delta h$; $x(i)$ is well logging values at depth $i \cdot \Delta h$; Δh is acquisition interval of well logging curves. According to fractal theory (Renata and Bary, 1999; Donald, 1997), measure $\mu(K,\varepsilon)$ relates scale $\varepsilon = M \cdot \Delta h$ as power law

$$\mu(K,\varepsilon) = A\varepsilon^{\alpha(K)} \tag{2}$$

where A is a constant; $\varepsilon = M \cdot \Delta h$ is scale; $\alpha(K)$ is a point K-wised constant, also, $\alpha(K)$ has intrinsic relation with fractal dimension of well logging curves at point K. $\alpha(K)$ delineates the sediment stacking pattern at K point and is called well logging singularity at points K. Assume $M=1$ in formula (1), so $\mu(K, \Delta h) = x(i) \cdot \Delta h$, and use it to replace μ in formula (2)

$$x(k) \cdot \Delta h = A \cdot (\Delta h)^{\alpha(K)} \tag{3}$$

(3)/(2)

$$\frac{x(k) \cdot \Delta h}{\mu(k,\varepsilon)} = \left(\frac{\Delta h}{M\Delta h}\right)^{\alpha(K)}$$

Then

$$x(k) = \frac{1}{\Delta h}\left(\frac{1}{M}\right)^{\alpha(K)} \cdot \mu(k,\varepsilon) \tag{4}$$

or

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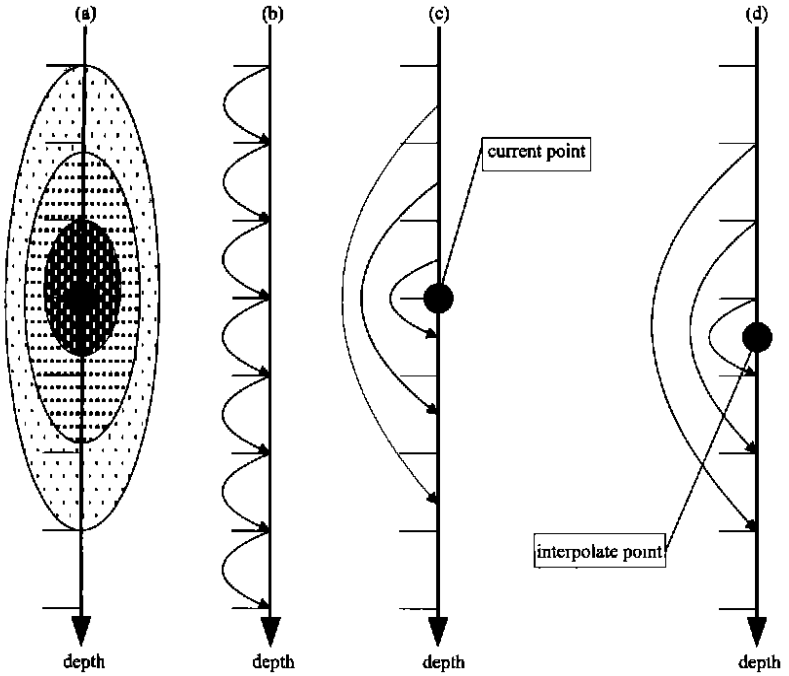


Figure 1. Well logging curve measurement definition demostration.

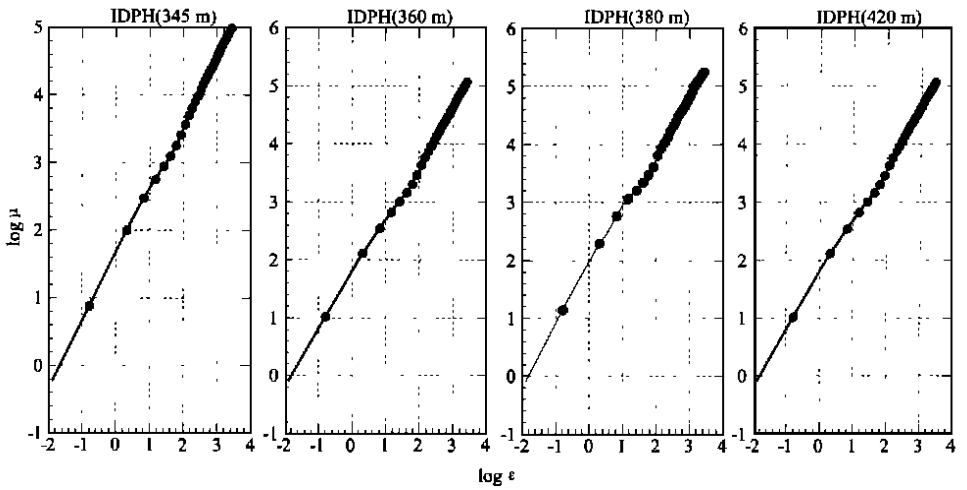


Figure 2. Measures of well logging curve vs scales show a linear line in dual logarithm coordinate system.

$$x(k) = \left(\frac{1}{M-1}\right)^{\alpha(K)} \sum_{i=K-\frac{M}{2},\ i\neq K}^{K+\frac{M}{2}} x(i) \tag{5}$$

Solve $X(K)$, by formula (5), we get formula (6)

$$x(k) = \left(\frac{1}{M}\right)^{\alpha(K)} \sum_{i=K-\frac{M}{2}}^{K+\frac{M}{2}} x(i) \tag{6}$$

Use formula (5) to recalculate known points for well logging correction, and use formula (6) to interpolate well logging values between two known adjacent points.

LOCAL SINGULARITY $\alpha(K)$ OF WELL LOGGING

From formula (2), $\log \mu(K, \varepsilon)$-$\log \varepsilon$ should be a linear relation in theory. But, well logging curves are affected by adjacent layers and borehole environment, therefore, the measures and scale maybe don't combine an exact line in dual logarithm coordinate system in real world. A least square error (LS) method is used to simulate a line passing through these of measures versus scales, and the slope of this simulated line is the best estimation of $\alpha(K)$ with the least square error significance.

Well logging curves of ODP Leg127, Hole 797C are used as test example. Figure 2 shows the LS linear regression results of curve IDPH at depths 345 m, 365 m, 380 m and 420 m mbsf (meter below sea floor) with M from 3, 5, 7, 71 (logging acquisition interval is 0.1 m, so the maximum scale length is 7.1 m) in dual logarithm coordinate system. It shows quite a good linear relation of measures and scale in dual logarithm coordinate system. The regression index and errors are displayed in Fig. 3.

How to choose the length of M depends on the vertical geometry of logging probes. If M is less, the corrected result will only reflect nearby concurrent points, and M should be in -

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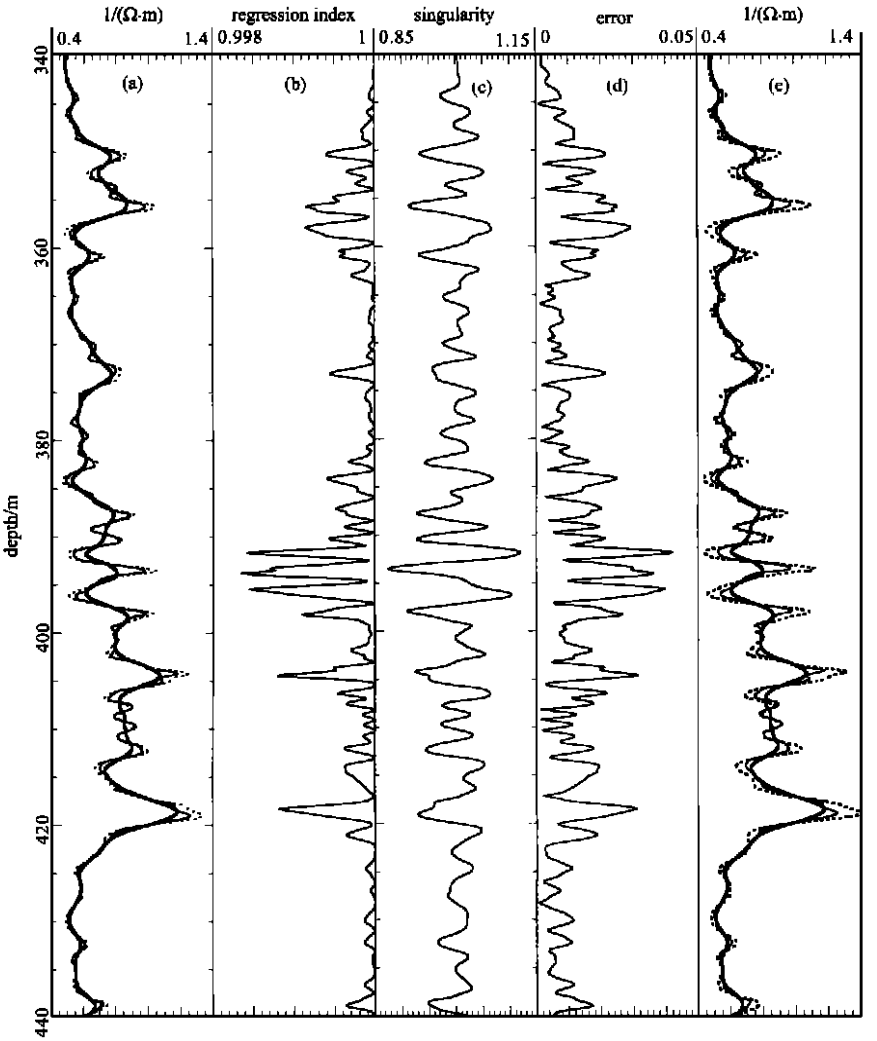


Figure 3. Deep and middle induced conductivity (IDPH, IMPH) of ODP Leg 127, Hole 797C original and corrected curves. (a), (e). Solid line for the IDPH and IMPH; Thick solid lines for the moving average results with 7.1 m window length; Dotted line for the fractal corrected result; (b). regression index for IMPH; (c). local singularity of IMPH; (d). local standard regression errors of IMPH.

creased if long-range effects are considered as experience based on many tests, $\alpha(K)$ is not sensitive to M changes in a wide range of M as in Fig. 2. It will not affect the slope of the line ($\alpha(K)$) if using a little less points in Fig. 2, because the regression line is quite stable and the regression index is exceptional, most of them is greater than 0.998 as shown in Fig. 3.

$\alpha(K)$ is the slope of measures vs scale in dual logarithm from formula (2) and Fig. 2. $\alpha(K)$ is the measure of concavity or convex in δ domain of point K. The slope should be 1 if the well logging curves keep stable in a series of expanding windows centered at concurrent point, herein the well logging curves need not to be corrected. If the well logging values increased is too fast compared with the scale expanding rate, the slope in Fig. 2 will be greater than 1 ($\alpha(K)>1$). The curve is with concave property in current point presenting that the stacking pattern around current point is with increasing model and current point is at the center of this expanding (sliding window) and concurrent well logging value is raised by adjacent layers and borehole environment with high values. So the correction is low down concurrent logging value. For the same reason, if $\alpha(K)<1$, logging curve is convex and the correct process will increase its value. So, $\alpha(K)$ represents the changing patterns in δ domain of current point. This correction method is reflected by the changing patterns in δ domain of interested point (Renata and Barry, 1999; Doveton, 1998; Van Wagoner, 1990).

IDPH, IMPH CORRECTION

The processed IDPH, IMPH results of Hole 797C, ODP Leg 127 are shown in Fig. 3. In Fig. 3a, the thin solid line is the IDPH curve; thick solid line, the moving average result of IDPH with 71 points sliding window length, and the thin dotted line, the fractal corrected result with the same window length of the moving average. The moving average makes maximum peaks a little lower and minimum peaks a little bit higher, this will lose the high frequency information. But, fractal correction method makes not only maximum points upper and minimum points lower in most cases to compensate the adjacent

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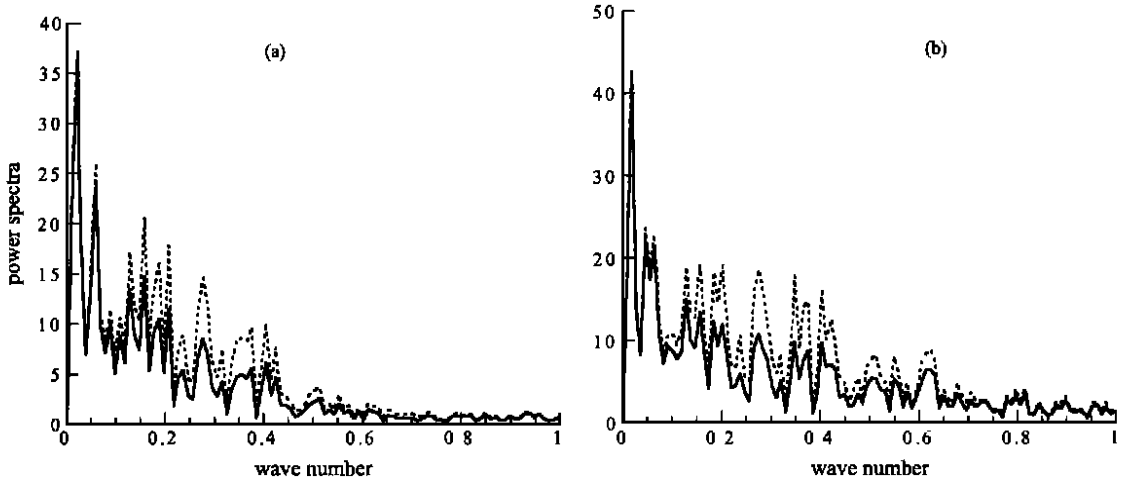


Figure 4. Power spectra of IDPH (a) and IMPH (b) against corrected curves (dotted lines).

effects, but also will make low points upper and upper points lower in some cases according to the stacking pattern nearby concurrent points.

Figure 3c shows the calculated local singularity $\alpha(K)$. $\alpha(K)$ fluctuates around 1. When $\alpha(K)>1$, fractal correction will low down the corresponded curves; when $\alpha(K)<1$, it will raise corresponded curves; when $\alpha(K)=1$, concurrent curve is not singular, so well logging curve need not to be corrected.

Figure 2b is the regression index in calculating $\alpha(K)$. All values are greater than 0.998, most of which is nearby 1. This means the defined measure in this paper is accordance with the fractal measure assumption. Figure 2d is the standard errors in calculation of $\alpha(K)$.

Corrected IMPH is shown in Fig. 3e. In it, thin solid line is IMPH curve, solid thick line is moving averaging with sliding window length of 71 points and the dotted line is fractal corrected result. IMPH correction shows the same results with IDPH.

DISCUSSION

The power spectra of IDPH and IMPH in Fig. 3a and 3e are estimated and plotted in Fig. 4a and 4b to compare the resolution message inside original and corrected curves. The fractal corrected well logging curve not only keep the original curve resolution message both in high and low frequency bands, but also enhance some component spectra value in middle frequency band as shown in Fig. 4. This will benefit in resolution and real data estimation. Herein, fractal correction method can be used in well logging correction.

The geometry characters of logging probe must be considered in well logging fractal correction. By adding weights in formula (5) and selecting M according to logging geometry, correction method could be improved and this needs future work.

This method could be also used in unevenly distributed situation by using average value in a window for measure calculation as shown in Fig. 1.

This method is programmed in 32 bits windows environment (WIN32) with friendly GUI. Interested reader can relate with the authors.

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